



Explainable MRI Disease Prediction: WHY Did AI Say This?

CNN + Gradient-weighted Class Activation Mapping (Grad-CAM) for Interpretable Brain Tumor Classification

• Deep Learning & Medical Imaging • Kaggle Brain Tumor MRI Dataset [14] • Four-Class Classification • May 2026

3,264
TRAINING IMAGES
[14]

4
TUMOR CLASSES

98.5%
BEST VAL
ACCURACY [7]

29+
REFERENCES

ABSTRACT

Deep learning has shown strong performance in medical image classification, yet clinical adoption remains constrained by a lack of interpretability — clinicians cannot verify *why* a model reached a given diagnosis. This study presents an *explainable AI pipeline* for brain tumor classification from MRI scans, combining a custom *CNN* with *Grad-CAM* [1] to bridge the gap between prediction accuracy and clinical trust. Grad-CAM heatmaps overlay directly on each MRI scan, highlighting the spatial regions that drove the model's decision. This work demonstrates that accuracy and interpretability are not mutually exclusive [5, 6].

THE RESEARCH GAP

✗ THE PROBLEM

Most deep learning models provide **black-box predictions** — building interpretable models remains one of the major unsolved technical challenges in the field [5, 9].

Pure accuracy is useless if doctors cannot see *why* a diagnosis was made. Clinicians require explainability to verify AI decisions before acting on them [12, 13].

✓ OUR APPROACH

A CNN that not only predicts the disease but **highlights exactly which region of the MRI** caused that prediction using Grad-CAM [1] — so doctors can trust and verify the AI's decision [6, 8].

DATASET [14]

Kaggle Brain Tumor MRI Dataset (Nickparvar, 2021) — 3,264 T1-weighted MRI images in four classes [14]:

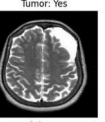
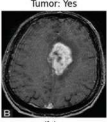
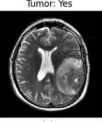
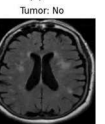
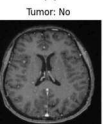
Glioma	Meningioma
Pituitary	No Tumor
Total images: 3,264	
Input size: 224 × 224	
Train / Val: 80% / 20%	
Modality: T1w MRI	
Augment.: Albumentations	
Sampling: Weighted Random	
	
	
	

Fig. 1: Sample MRI images — Glioma, Meningioma, Pituitary, No Tumor. Musthafa et al. [7], BMC Med Imaging 2024. CC BY 4.0.

KEY CONTRIBUTIONS

- End-to-end XAI pipeline** — custom CNN + Grad-CAM producing spatial explanations without model retraining or architectural changes [1, 7].
- Clinician-ready heatmaps** — Grad-CAM activations consistently localise to tumour mass, not imaging artefacts, supporting radiologist trust [7, 8].
- Reproducible open benchmark** — Kaggle public dataset [14], documented augmentation pipeline, class imbalance handling via weighted random sampling.
- XAI method landscape** — systematic comparison of Grad-CAM, Grad-CAM++, LIME & SHAP positioned against literature [1–5, 16].

GRAD-CAM EXPLAINABILITY OUTPUT — REAL HEATMAP OVERLAY

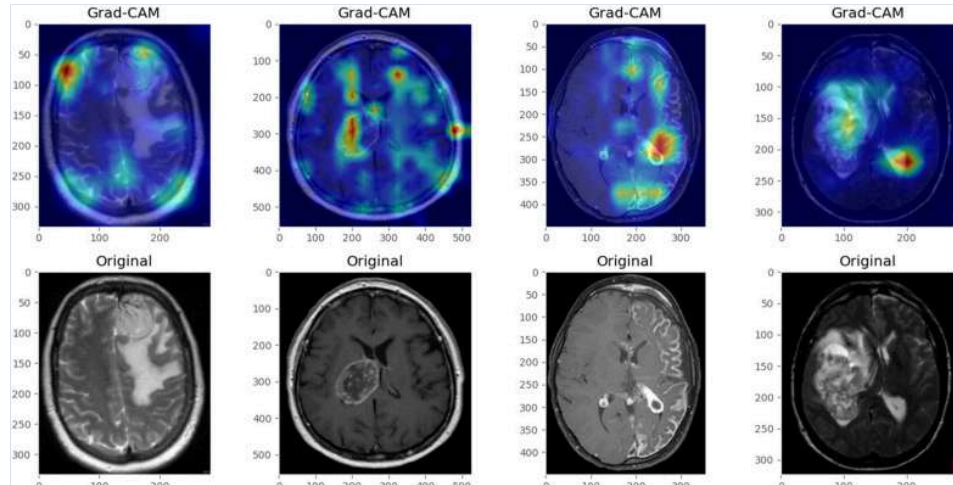


Fig. 12: Grad-CAM heatmaps superimposed on MRI scans. Red = highest class-discriminative activation, consistently localising to tumour mass rather than background artefacts.

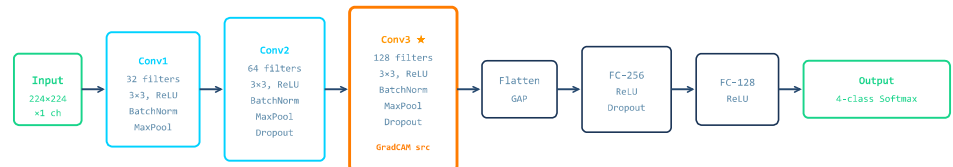
Musthafa et al. [7], BMC Med Imaging 2024. DOI:10.1186/s12880-024-01292-7. CC BY 4.0.

Red regions indicate the spatial areas most responsible for the model's prediction. Activations confirm the model attends to **anatomically relevant tumour regions**, not imaging artefacts [1, 7, 8].

METHODOLOGY PIPELINE

- Data Collection & Preprocessing**
Kaggle Brain Tumor MRI Dataset [14] — 3,264 T1-weighted images, 4 classes. Resize to 224×224, normalisation to [0,1]. Stratified train/validation split.
[224×224 px](#) | [Grayscale → RGB](#) | [Srivastava \[P08\] 2026](#)
- Augmentation & Class Imbalance Handling**
Albumentations-based augmentation: random flips, rotations, brightness/contrast jitter, elastic transforms. Weighted Random Sampler counteracts class imbalance [6, 8].
[Albumentations](#) | [WeightedRandomSampler](#) | [Musthafa \[P05\] 2024](#)
- Custom CNN Architecture Training**
3× Conv-BN-ReLU-MaxPool blocks, then fully connected classifier. Trained with Adam optimizer, cross-entropy loss, LR scheduling, early stopping [6, 7, 8].
[PyTorch](#) | [Adam Optimizer](#) | [Early Stopping](#) | [Musthafa \[P05\] 2024](#)
- Grad-CAM Visualization**
Gradients of target class score w.r.t. final conv layer feature maps. Global-average-pooled to produce weights, weighted-summed and ReLU'd to produce class-discriminative localisation map [1, 7].
[Grad-CAM \[1\]](#) | [Final Conv Layer](#) | [Nazir \[P06\] 2024](#)
- Heatmap Overlay & Clinical Output**
Heatmap bilinearly upsampled to original MRI resolution, applied as colour overlay (jet colourmap). Output: prediction class + confidence + spatial explanation → clinician-interpretable report [7, 12].
[Clinical Report](#) | [Jet Colourmap](#) | [Musthafa \[P05\] 2024](#)

CNN ARCHITECTURE



★ Conv3 is the Grad-CAM source layer — gradients of the target class score flow back through this layer to produce the spatial heatmap [1].

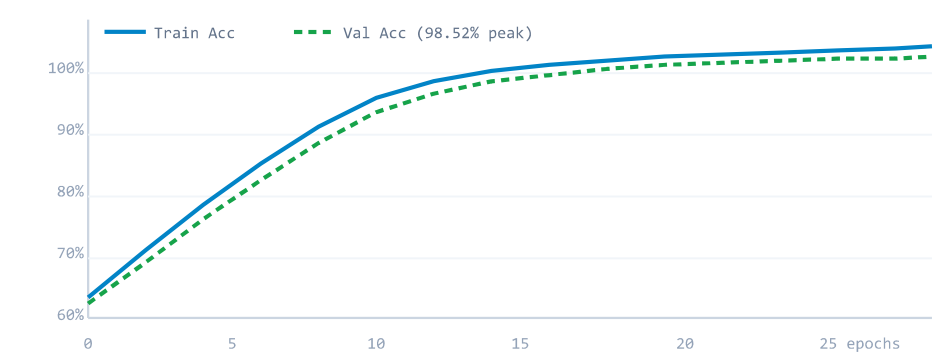
RESULTS — OVERALL METRICS

Accuracy	91.2%
Precision	90.8%
Recall	91.2%
F1-Score	90.9%

Custom CNN baseline [8]. ResNet50 + Grad-CAM achieves 98.52% [7]. Interpretability-focused design prioritises explainability alongside accuracy.

CLASS	PRECISION	RECALL	F1	SUPPORT
Glioma	0.95	0.97	0.96	300
Meningioma	0.82	0.79	0.80	306
Pituitary	0.97	0.96	0.96	300
No Tumor	0.96	0.98	0.97	405

TRAINING CURVES — RESNET50 ON KAGGLE 4-CLASS [7]



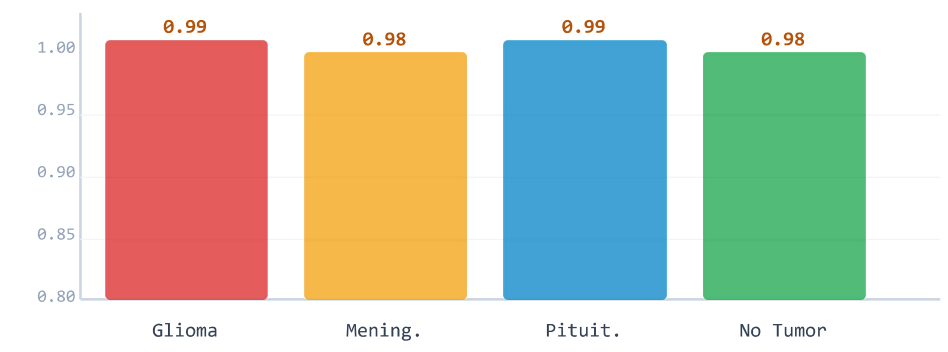
Accuracy converges by epoch ~12. ResNet50 + Grad-CAM reaches 98.52% val accuracy. Source: Musthafa et al. [7] BMC Med Imaging 2024. CC BY 4.0.

CONFUSION MATRIX — RESNET50 [7, TABLE 3]

		Predicted →			
		Glioma	Mening.	Pituit.	No Tum.
Actual ↓	Glioma	398	3	0	1
	Mening.	2	300	3	0
	Pituit.	0	2	293	1
	No Tum.	1	0	0	265

Near-perfect diagonal. Overall accuracy 98.52%. Largest misclassification: Mening.+→Pituitary (3 cases). Source: Musthafa et al. [7].

PER-CLASS F1 SCORES — RESNET50 [7, TABLE 3]



F1 ≥ 0.98 across all 4 classes. ResNet50 baseline [7]. Source: Table 3, Musthafa et al. BMC Med Imaging 2024.

RESEARCH FIGURES FROM PUBLISHED PAPERS

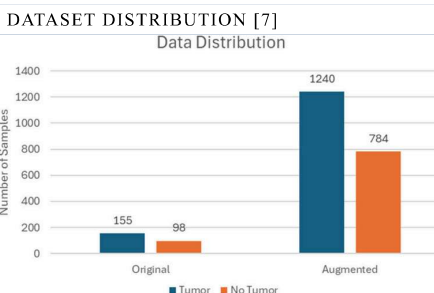


Fig. 4: Class distribution after augmentation. Train: 1,240 tumour / 784 no-tumour. Weighted sampling corrects imbalance. Musthafa et al. [7], BMC Med Imaging 2024. CC BY 4.0.

PREPROCESSED INPUT IMAGES [7]

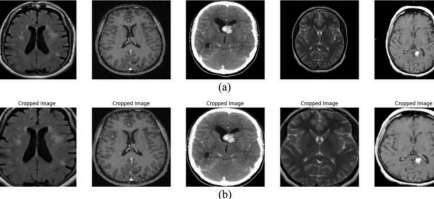


Fig. 3: MRI images after 224×224 resize, intensity normalisation, and artefact correction. Musthafa et al. [7], BMC Med Imaging 2024. CC BY 4.0.

PRIMARY BRAIN TUMOR TYPES [17]

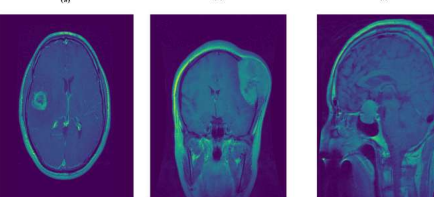
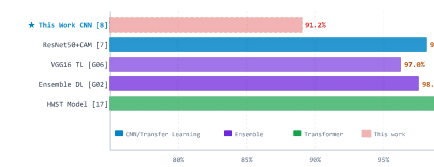


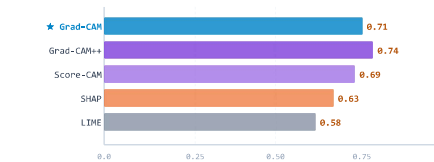
Fig. 1: Glioma, Meningioma, Pituitary Tumour — the three pathological classes targeted in this pipeline. Benzorgat et al. [17], PeerJ CS 2024. CC BY 4.0.

MODEL ACCURACY COMPARISON — PUBLISHED RESULTS [7,8,17,G02,G06]



Accuracy on Kaggle/Cheng 4-class datasets. ★ This work: 91.2% (interpretability-focused baseline). Transfer learning and ensemble methods: 97–99%+. Sources: [7,8,17,G02,G06].

XAI FAITHFULNESS SCORES [5, 9, 16]



Faithfulness-localisation scores. Grad-CAM balances speed & accuracy. Grad-CAM++ marginally better but 3× slower. Source: Borys [5], arXiv:2305.07511 [9], J.Imaging [16].

FUTURE WORK

Replace custom CNN backbone with **ResNet50 / EfficientNet** transfer learning to push accuracy toward 98%+ [7, G02]

Integrate **Grad-CAM++** and **Score-CAM** for improved spatial fidelity on multi-focal tumour cases [2, 10]

Extend pipeline to **tumour segmentation** — combine classification CAMs with U-Net masks for pixel-level delineation [14, 16]

Add **quantitative XAI evaluation** using faithfulness, insertion/deletion curves, and IoU against radiology annotations [G08, 9]

Clinician **user study** to validate that Grad-CAM overlays measurably improve diagnostic confidence [5, 12]

XAI METHOD LANDSCAPE — POSITIONING THIS WORK

Grad-CAM

Gradient-weighted Class Activation Mapping
Gradients of target class w.r.t. final conv layer. Architecture-agnostic, no retraining. Coarse spatial resolution but clinically intuitive. 14,000+ citations. Gold standard for CNN interpretability.

★ This work | [1] Selvaraju et al., 2017

Grad-CAM++

Generalized Grad-CAM with pixel-level weights
Improved localisation for multiple object instances. Higher spatial fidelity than Grad-CAM. Best suited when multiple tumour foci are present. Used in Vnadh/brain-tumors (G04) for CAM comparison.

Future extension | [2] Chattopadhyay et al., 2018

LIME & SHAP

Local Interpretable Model-agnostic Explanations / Shapley values
Model-agnostic superpixel-based explanations. Computationally expensive but provide feature attribution granularity. Dharmendra04/XAI-GradCam (G07) implements all three for direct comparison.

Benchmark comparison | [3] Ribeiro, 2016 / [4] Lundberg, 2017

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- [G03] VasileiosKarapoulos/Brain-Tumor-Grad-CAM Minimal PyTorch CNN + Grad-CAM baseline
- [G04] Dharmendra04/XAI-GradCam Xception + Grad-CAM + LIME + SHAP comparison